

Balancing Background Lighting and Task Lights under Dynamic Occupancy: A Theoretical Energy-Saving Study

ABSTRACT

Traditional building design is typically based on full-load occupancy, yet actual workplace occupancy is increasingly variable. This mismatch leads to over-illumination and waste in underutilized spaces. While background lighting focuses on macro-scale ambient illumination, task lights address micro-scale individual needs. Integrating these two scales offers a promising approach to occupancy-lighting mismatches. To address this gap, this paper establishes a Balanced Light Control Framework to investigate their combined energy-saving potential while maintaining visual comfort.

This study constructs a 3D model of an office laboratory in Singapore and evaluates it under a parametric simulation matrix. The potential is explored across three representative sky conditions and five occupancy levels derived from the range of observed occupancy peaks. To better represent occupants' variability in the case study, 10 random sitting patterns were generated for each density level, yielding 150 unique simulation scenarios. For each sky-occupancy scenario, an objective function was formulated on total lighting energy consumption and penalty terms associated with unmet illuminance and uniformity requirements defined by SS 531. The ceiling light group levels and task light levels were the parameters to be optimized. A genetic algorithm (via Galapagos component) was then used to minimize this objective function as well as identify the minimum-energy lighting configuration for each scenario.

Preliminary results show that the optimized strategy consistently achieved the lowest average lighting power across all tested occupancy conditions. Compared with a rule-based combined lighting strategy with preset ceiling and task light levels, it reduced average lighting power by approximately 15–34%, and by about 40–66% relative to the ceiling-light-only baseline. These results suggest that the main benefit is not only in introducing task lights, but in coordinating them with background lighting according to changing occupancy conditions. This performance ceiling can serve as a case-based reference for evaluating occupancy-centric lighting strategies in similar office contexts.

KEYWORDS

Occupancy-centric lighting control, Energy saving, Genetic algorithm, Visual comfort, Task lights